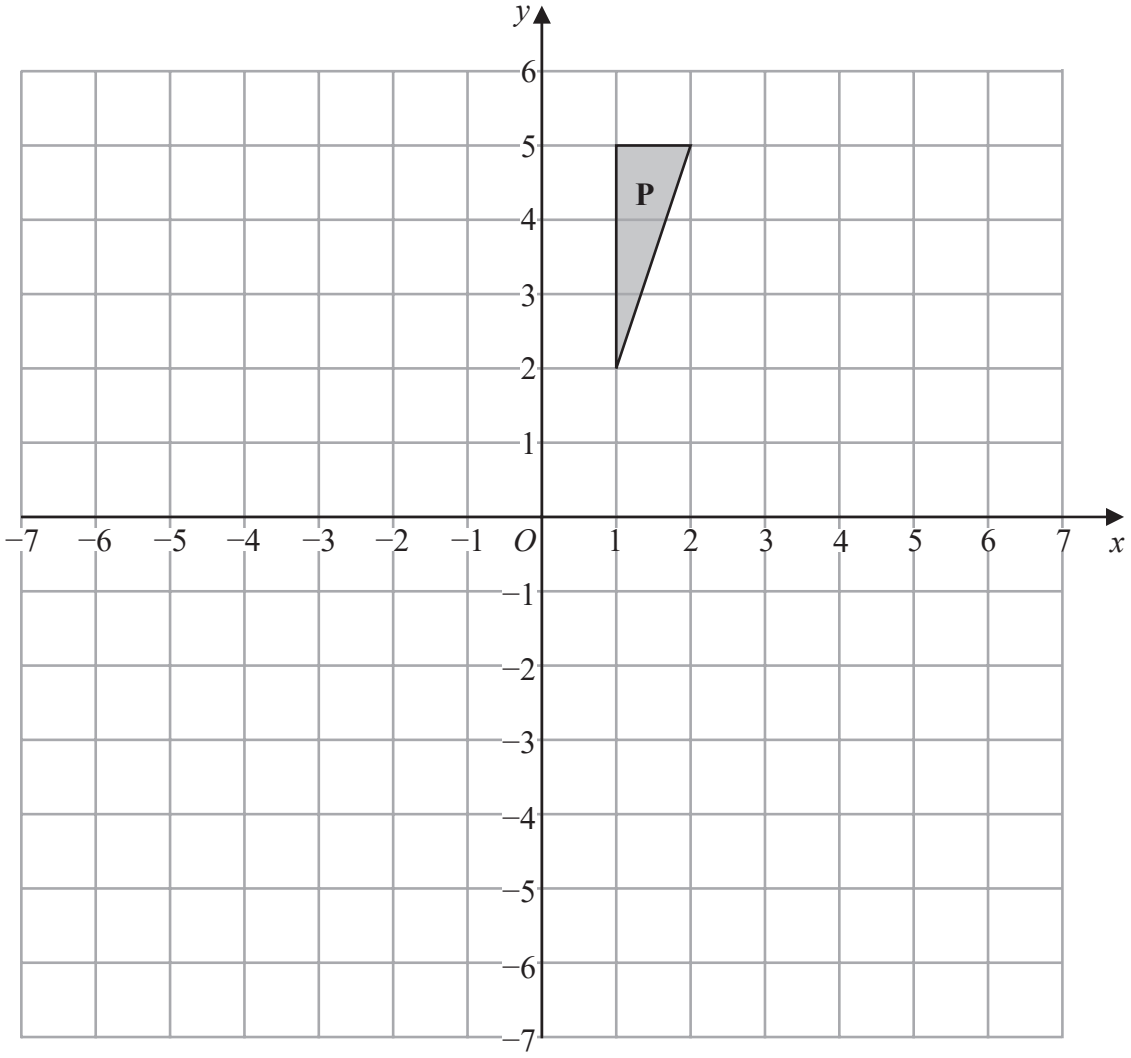
11 The diagram shows a triangle **P** on a grid.



Triangle **P** is rotated 180° about (0, 0) to give triangle **Q**.

Triangle **Q** is translated by $\begin{pmatrix} 5 \\ -2 \end{pmatrix}$ to give triangle **R**.

(a) Describe fully the single transformation that maps triangle **P** onto triangle **R**.

180° rotation: P(x,y) → (−x,−y).

Then translation (5,−2): R(x,y)=(5−x,−2−y).

This is a 180° rotation about (2.5, −1).

3 marks total
Find R from P
180° rotation
Centre (2.5,−1)

(3)

Under the transformation that maps triangle **P** onto triangle **R**, the point A is invariant.

(b) Write down the coordinates of point A.

The invariant centre is A = (2.5, −1).

(..... , ...)

(1)

1 mark
A = (2.5, −1)

(Total for Question 11 is 4 marks)

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